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Maximum likelihood estimation in a class of nonregular cases

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SUMMARY

We consider maximum likelihood estimation of the parameters of a probability density which is zero for $x < \theta$ and asymptotically $\alpha c(x - \theta)^{\alpha-1}$ as $x \downarrow \theta$. Here θ and other parameters, which may or may not include α and c , are unknown. The classical regularity conditions for the asymptotic properties of maximum likelihood estimators are not satisfied but it is shown that, when $\alpha > 2$, the information matrix is finite and the classical asymptotic properties continue to hold. For $\alpha = 2$ the maximum likelihood estimators are asymptotically efficient and normally distributed, but with a different rate of convergence. For $1 < \alpha < 2$, the maximum likelihood estimators exist in general, but are not asymptotically normal, while the question of asymptotic efficiency is still unsolved. For $\alpha \leq 1$, the maximum likelihood estimators may not exist at all, but alternatives are proposed. All these results are already known for the case of a single unknown location parameter θ , but are here extended to the case in which there are additional unknown parameters. The paper concludes with a discussion of the applications in extreme value theory.

Some key words: Extreme value theory; Maximum likelihood; Nonregular estimation; Stable distribution; Weibull distribution.

1. INTRODUCTION

It is well known that, if the support of a probability density depends on an unknown parameter, then the classical regularity conditions for maximum likelihood estimation are not satisfied. In some cases the maximum likelihood estimators exist and have the same asymptotic properties as in regular cases. In other cases they may exist but not be asymptotically efficient or normally distributed, and in still other cases they may not exist at all, at least not as solutions of the likelihood equations.

The case of a single unknown location parameter

$$f(x; \theta) = f_0(x - \theta) \quad (\theta < x < \infty); \quad f_0(x) \sim \alpha c x^{\alpha-1} \quad (1.1)$$

as $x \downarrow 0$ ($\alpha > 0, c > 0$), subject to certain smoothness conditions, is well documented. For $\alpha > 2$ the Fisher information is finite and the maximum likelihood estimators have the same asymptotic properties as in regular cases. This is stated by Woodroffe (1972, Proposition 1.1), based on results of Le Cam (1970); the result is also implicit in Dawid's (1970) proof of the asymptotic normality of posterior distributions in this case. For $\alpha = 2$ the maximum likelihood estimators are asymptotically normal (Woodroffe, 1972; Akahira, 1975a) and efficient (Weiss & Wolfowitz, 1973), but the order of convergence is $O\{(n \log n)^{\frac{1}{2}}\}$ instead of the usual $O(n^{\frac{1}{2}})$. For $1 < \alpha < 2$ the maximum likelihood estimators have a nonnormal limiting distribution with order of convergence $O(n^{1/\alpha})$.

(Woodroffe, 1974). Asymptotic efficiency is an open question, though Akahira (1975a, b) showed that no estimator has a larger order of convergence. When $\alpha \leq 1$ the likelihood function has no local maximum but is globally maximized at the sample minimum. The sample minimum itself is a consistent estimator with order of convergence $O(n^{1/\alpha})$, and Akahira's results show that, for $0 < \alpha < 2$, this order of convergence cannot be improved. When $\alpha = 1$ a result of Weiss (1979) shows that the sample minimum has a property of asymptotic sufficiency, but I have not succeeded in applying Weiss's argument for any other value of α .

In the present paper these results are extended to distributions in which there are other unknown parameters as well as θ . Some examples which fall within our framework are as follows.

- (i) Three-parameter Weibull,

$$f(x; \theta, \alpha, \beta) = \alpha\beta(x-\theta)^{\alpha-1} \exp\{-\beta(x-\theta)^\alpha\} \quad (\theta < x < \infty, \alpha > 0, \beta > 0). \quad (1.2)$$

- (ii) Three-parameter gamma,

$$f(x; \theta, \alpha, \beta) = \beta^\alpha(x-\theta)^{\alpha-1} \exp\{-\beta(x-\theta)\}/\Gamma(\alpha) \quad (\theta < x < \infty, \alpha > 0, \beta > 0). \quad (1.3)$$

- (iii) Three-parameter beta, which is a beta distribution with an unknown scale parameter, which can be recast in our framework by defining X to be a random variable on $\theta < x < \infty$ with $1 - \exp(-X + \theta)$ beta distributed; thus

$$f(x; \theta, \alpha, \beta) = B(\alpha, \beta) \{1 - e^{-(x-\theta)}\}^{\alpha-1} e^{-\beta(x-\theta)} \quad (\theta < x < \infty, \alpha > 0, \beta > 0). \quad (1.4)$$

- (iv) Three-parameter log gamma, which arises when $\log(X - \theta + 1)$ has a gamma distribution; thus

$$f(x; \theta, \alpha, \beta) = \beta^\alpha \{\log(x - \theta + 1)\}^{\alpha-1} (x - \theta + 1)^{-\beta-1} / \Gamma(\alpha) \quad (\theta < x < \infty, \alpha > 0, \beta > 0). \quad (1.5)$$

Our results also cover certain instances of the Box-Cox transformation family but we shall not consider these explicitly.

Of these four examples, the Weibull has been studied the most intensively. For all four, when $\alpha < 1$ the density tends to infinity as $x \downarrow \theta$ so that, unless the range $\alpha < 1$ is excluded, the likelihood function always tends to infinity along some path in the likelihood space as θ tends to the sample minimum. Therefore it is necessary to distinguish between global and local maxima of the likelihood. In this paper, by the maximum likelihood estimator we shall always mean a local maximum, thus satisfying the likelihood equations. A second point, which applies to all four examples but may not be true in general, is that for $\alpha < 1$ the density is J-shaped, so there cannot exist a maximum likelihood estimator for which $\alpha < 1$. In particular, if the true value of α is less than 1, maximum likelihood estimators either do not exist at all or are inconsistent.

Harter & Moore (1965) described an iterative procedure for finding maximum likelihood estimators for the Weibull and gamma distributions with possibly censored data. In cases where maximum likelihood estimators do not exist, they proposed an ad hoc modification based on treating the smallest observation as if it were censored. Rockette, Antle & Klimko (1974) showed for the Weibull distribution that, if a local maximum of the likelihood function exists, then there is a second solution of the likelihood equations which is a saddlepoint. Their result shows that, in finding maximum

likelihood estimators one must take care to ensure that a solution of the likelihood equations really is a local maximum. Other relevant references are Cohen (1965) and Lemon (1975). The Weibull distribution may be reparameterized as the generalized extreme value distribution (Jenkinson, 1955) for which modern algorithms are available (NERC, 1975; Prescott & Walden, 1980, 1983).

Thus there is a fair-sized literature on finding maximum likelihood estimators of the Weibull distribution, but their asymptotic properties are largely unexplored. It is easily checked that, when $\alpha > 2$, the Fisher information matrix is finite, and it is widely assumed that the classical properties hold in this case. For $\alpha \leq 2$ the Fisher information for θ is infinite, so the classical results are certainly not valid.

In this paper it is confirmed that the classical results hold when $\alpha > 2$, and the case $\alpha \leq 2$ is studied in detail. The surprising result is that, in this case, estimation of θ and the other parameters, denoted by ϕ , in general a vector, are asymptotically independent: each of the maximum likelihood estimators of θ and ϕ has the same asymptotic distribution when the other is unknown as when the other is known, and we are also able to show that these asymptotic distributions are independent. For $1 < \alpha < 2$ we prove the existence of a consistent sequence of maximum likelihood estimators as the sample size tends to infinity, while for $\alpha < 1$ no consistent maximum likelihood estimators exist. We also propose efficient alternatives to maximum likelihood which, in particular, cover the cases where maximum likelihood estimators do not exist.

Cheng & Amin (1983) have also studied the asymptotic properties of maximum likelihood estimators for the Weibull and gamma cases, though their Theorem 2 is less extensive than our results in §§3 and 4 and their proofs are unpublished. On the other hand they introduce a new estimator, the ‘maximum product of spacings’ estimator, which deserves to be studied further. Johnson & Haskell (1983) prove consistency of the maximum likelihood estimator for the three-parameter Weibull with $\alpha > 1$, and present Monte Carlo results which indicate that, even in the ‘regular’ case $\alpha > 2$, the asymptotic normality of the estimators is approached only slowly.

Our main results require a long list of assumptions, which are stated in §2. The remainder of the paper is organized so that statements of the main results appear at the beginning of each section, and may be understood without reference to the proofs. Results of a purely technical nature are stated as lemmas and contained within the proofs of the theorems, though Lemmas 6 and 7 may be of independent interest.

2. ASSUMPTIONS

We consider probability densities of the form

$$f(x; \theta, \phi) = (x - \theta)^{\alpha-1} g(x - \theta; \phi) \quad (\theta < x < \infty), \quad (2.1)$$

where θ and ϕ , the latter a vector, are unknown parameters and g tends to a constant αc as $x \downarrow \theta$. We assume $\alpha \equiv \alpha(\phi)$ is a twice continuously differentiable function of ϕ with $0 < \alpha < \infty$. This formulation allows both the case when α is a component of ϕ and where it is a known constant. Similarly $c \equiv c(\phi)$. For our four examples the function g is given by:

- (i) Weibull, $g(x; \alpha, \beta) = \alpha \beta \exp(-\beta x^\alpha);$
- (ii) gamma, $g(x; \alpha, \beta) = \beta^\alpha \exp(-\beta x) / \Gamma(\alpha);$
- (iii) beta, $g(x; \alpha, \beta) = B(\alpha, \beta) \{(1 - e^{-x})/x\}^{\alpha-1} e^{-\beta x};$
- (iv) log gamma, $g(x; \alpha, \beta) = \beta^\alpha \{(\log(1+x))/x\}^{\alpha-1} (1+x)^{-\beta-1} / \Gamma(\alpha).$

In general, we assume θ is real-valued and ϕ takes values in a set $\Phi \subseteq \mathbb{R}^p$ for some $p \geq 1$. We adopt the convention that subscripts, ϕ_1, ϕ_2 , etc., are used to denote elements of Φ , whilst superscripts, ϕ^1, ϕ^2 , etc., denote components of a particular ϕ . Thus ϕ_j^i ($i = 1, \dots, p$) is the i th component of ϕ_j . The symbol $|\phi|$ denotes the L^2 norm of ϕ .

The detailed assumptions are as follows.

Assumption 1. All second-order partial derivatives of $g(x; \phi)$ exist and are continuous in $0 < x < \infty, \phi \in \Phi$. Moreover $c(\phi) = \alpha^{-1} \lim_{x \downarrow 0} g(x; \phi)$ as $x \downarrow 0$ exists, is positive and finite for each ϕ , and is twice continuously differentiable as a function of ϕ .

Assumption 2. If ϕ_1 and ϕ_2 are distinct elements of Φ then the set

$$\{x: f(x; 0, \phi_1) \neq f(x; 0, \phi_2)\}$$

has positive Lebesgue measure.

Assumption 3. For some fixed $\delta^* \geq 0$ and for each $\eta > 0, \phi_1 \in \Phi$, we have

$$\log g(x - \varepsilon; \phi) - \log g(x - \varepsilon_1; \phi_1) < H_\eta(x; \phi_1),$$

whenever $|\varepsilon| < \eta, |\varepsilon_1| < \eta, |\phi - \phi_1| < \eta, x - \varepsilon > \delta^*, x - \varepsilon_1 > \delta^*$ and the function H_η satisfies

$$\lim_{\eta \rightarrow 0} \int_0^\infty H_\eta(x + y; \phi_1) f(x; 0, \phi_0) dx = 0$$

for each $y \geq 0, \phi_0 \in \Phi$.

Assumption 4. There exists a fixed increasing sequence of compact subsets $\{K_m, m \geq 1\}$ of $\mathbb{R}^{p+1}$ and a fixed constant δ' such that $\bigcup_m K_m = \mathbb{R} \times \Phi$, and, for $\theta, \theta_0, \phi, \phi_0$ satisfying $\theta < \theta_0 - \delta', \phi, \phi_0 \in \Phi, (\theta, \phi) \notin K_m$, we have

$$\log f(x; \theta, \phi) - \log f(x; \theta_0, \phi_0) < H_m^0(x; \theta_0, \phi_0) \quad (x > \theta_0),$$

where

$$-\infty \leq \limsup_{m \rightarrow \infty} \int_{\theta_0}^\infty H_m^0(x; \theta_0, \phi_0) f(x; \theta_0, \phi_0) dx < 0.$$

Assumption 5. There exists a fixed increasing sequence of compact subsets $\{K'_m, m \geq 1\}$ of Φ with $\bigcup_m K'_m = \Phi$, and $\delta' > 0$ a fixed constant, such that for $\phi \in \Phi - K'_m, \phi_0 \in \Phi, |\varepsilon| < \delta', \max(0, \varepsilon) < x < \infty$ and some $\eta' > 0$, we have

$$\eta' \alpha(\phi) + \{\alpha(\phi) - \alpha(\phi_0)\} \log x + \log g(x - \varepsilon; \phi) - \log g(x; \phi_0) < H'_m(x; \phi_0),$$

where

$$-\infty \leq \limsup_{m \rightarrow \infty} \int_0^\infty H'_m(x; \phi_0) f(x; 0, \phi_0) dx < 0.$$

Assumption 6. If E_ϕ denotes expectation with respect to $f(\cdot; 0, \phi)$ then for each $\phi \in \Phi$ ($i, j = 1, \dots, p$):

(a)

$$E_\phi \left\{ \left(\frac{\partial}{\partial \phi^i} \right) \log f(X; 0, \phi) \right\} = 0,$$

$$\begin{aligned} E_\phi \left\{ \left(\frac{\partial}{\partial \phi^i} \right) \log f(X; 0, \phi) \left(\frac{\partial}{\partial \phi^j} \right) \log f(X; 0, \phi) \right\} &= -E_\phi \left\{ \left(\frac{\partial^2}{\partial \phi^i \partial \phi^j} \right) \log f(X; 0, \phi) \right\} \\ &= m_{ij}(\phi); \end{aligned}$$

(b) if $\alpha > 1$, then

$$\begin{aligned} E_\phi \left\{ \left(\frac{\partial}{\partial x} \right) \log f(X; 0, \phi) \right\} &= 0, \\ -E_\phi \left\{ \left(\frac{\partial}{\partial x} \right) \log f(X; 0, \phi) \left(\frac{\partial}{\partial \phi^i} \right) \log f(X; 0, \phi) \right\} &= E_\phi \left\{ \left(\frac{\partial^2}{\partial x \partial \phi^i} \right) \log f(X; 0, \phi) \right\} \\ &= m_{i0}(\phi) = m_{0i}(\phi); \end{aligned}$$

(c) if $\alpha > 2$, then

$$\begin{aligned} E_\phi [\{(\partial/\partial x) \log f(X; 0, \phi)\}^2] &= -E_\phi \{(\partial^2/\partial x^2) \log f(X; 0, \phi)\} \\ &= m_{00}(\phi) > 0. \end{aligned}$$

It is part of the assumption that all these expectations are finite.

Assumption 7. If $h(x; \phi)$ is any of $(\partial^2/\partial x \partial \phi^i) \log g(x; \phi)$ or $(\partial^2/\partial \phi^i \partial \phi^j) \log g(x; \phi)$, then, as $\theta \rightarrow \theta_0, \phi \rightarrow \phi_0$,

$$E_0 |h(X - \theta; \phi) - h(X - \theta_0; \phi_0)| \rightarrow 0,$$

where E_0 is with respect to $f(\cdot; \theta_0, \phi_0)$. If $\alpha(\phi_0) > 2$, we require the same of $h(x; \phi) = (\partial^2/\partial x^2) \log g(x; \phi)$.

Assumption 8. For each $\varepsilon > 0, \delta > 0$, there exists a function $h_{\varepsilon, \delta}$ such that

$$|(\partial^2/\partial x^2) \log g(x; \phi)| < \varepsilon/x^2 + h_{\varepsilon, \delta}(y; \phi_0)$$

whenever $|\phi - \phi_0| < \delta, |x - y| < \delta$, and $h_{\varepsilon, \delta}$ satisfies

$$\int_0^\infty h_{\varepsilon, \delta}(x; \phi_0) f(x; 0, \phi_0) dx < \infty.$$

Assumption 9. For each $\delta > 0, \phi \in \Phi$,

$$\int_\delta^\infty \{(\partial/\partial x) \log g(x; \phi)\}^2 f(x; 0, \phi) dx < \infty.$$

We now make some remarks on the assumptions. The key assumptions are Assumptions 1 and 6; the rest are there for technical reasons. Assumptions 2–5 are similar to, but necessarily more complicated than, the classical assumptions of Wald (1949). In Assumption 3 we may have $\delta^* = 0$ or $\delta^* > 0$; in the Weibull case, Assumption 3 is true with $\delta^* > 0$ for all α but with $\delta^* = 0$ only for $\alpha > 1$. The distinction is reflected in the statement of Theorem 2 below. Assumptions 7 and 8 are needed for Lemma 4 in §3, and Assumption 9 is taken from Woodroffe (1972, 1974). Assumption 1 could be weakened to $g(x; \phi)$ slowly varying at $x = 0$ for each ϕ , but at the cost of an increase in technical detail.

For all our examples, Assumptions 1–6 and 9 are straightforward, if somewhat tedious, to check. For examples (ii)–(iv), $\log g$ and its partial derivatives are bounded in x for each ϕ , so Assumptions 7 and 8 are easy to check as well. For the Weibull distribution we have

$$\begin{aligned} \log g(x; \alpha, \beta) &= \log \alpha + \log \beta - \beta x^\alpha, \\ \frac{\partial}{\partial x} \log g(x; \alpha, \beta) &= -\alpha \beta x^{\alpha-1}, \quad \frac{\partial^2}{\partial x^2} \log g(x; \alpha, \beta) = -\alpha(\alpha-1) \beta x^{\alpha-2}, \end{aligned}$$

which are bounded as $x \downarrow 0$ respectively for $\alpha > 0, \alpha > 1, \alpha > 2$; Assumption 7 follows easily. To satisfy Assumption 8, given $0 < \alpha_1 < \alpha_0 < \alpha_2$ and $\delta > 0$, there exists K such that

$$\left| \frac{\partial^2}{\partial x^2} \log g(x; \alpha, \beta) \right| < \begin{cases} Kx^{\alpha_1-2} & (x \leq 1), \\ Kx^{\alpha_2-2} & (x \geq 1), \end{cases}$$

and all (α, β) satisfying $\alpha_1 < \alpha < \alpha_2, |\beta - \beta_0| < \delta$. Let $x_\varepsilon = (\varepsilon/K)^{1/\alpha}$ which we assume to be ≤ 1 . Then

$$|(\partial^2/\partial x^2) \log g(x; \alpha, \beta)| < \varepsilon/x^2 + h'_{\varepsilon, \delta}(x; \alpha_0, \beta_0),$$

where

$$h'_{\varepsilon, \delta}(x; \alpha_0, \beta_0) = \begin{cases} 0 & (x \leq x_\varepsilon), \\ Kx^{\alpha_1-2} & (x_\varepsilon \leq x \leq 1), \\ Kx^{\alpha_2-2} & (x \geq 1). \end{cases}$$

This function is bounded as $x \rightarrow 0$. If we define $h_{\varepsilon, \delta}(y; \alpha_0, \beta_0)$ to be

$$\sup \{h'_{\varepsilon, \delta}(x; \alpha_0, \beta_0) : |x - y| < \delta\},$$

Assumption 8 is satisfied.

3. EXISTENCE AND UNIQUENESS OF CONSISTENT ESTIMATORS

From now on we shall let θ_0, ϕ_0 denote the true values of θ and ϕ . The letters α and c , unless indicated otherwise, will always denote $\alpha(\phi_0)$ and $c(\phi_0)$. Let $X_1, \dots, X_n$ denote a sample of independent observations from the common density $f(\cdot; \theta_0, \phi_0)$. Let L_n denote the log likelihood divided by n , that is

$$L_n(\theta, \phi) = n^{-1} \sum_{i=1}^n \log f(X_i; \theta, \phi).$$

The order statistics will be denoted $X_{n,1} < \dots < X_{n,n}$; note that $L_n(\theta, \phi)$ is defined only for $\theta < X_{n,1}$.

The maximum likelihood estimator, when it exists, will be denoted by $(\hat{\theta}_n, \hat{\phi}_n)$ and thus satisfies

$$\partial L_n(\hat{\theta}_n, \hat{\phi}_n)/\partial \theta = 0, \quad \partial L_n(\hat{\theta}_n, \hat{\phi}_n)/\partial \phi^i = 0 \quad (i = 1, \dots, p).$$

For the special case when $\theta = \theta_0$ is known, let $\bar{\phi}_n \equiv \bar{\phi}_n(\theta_0)$ denote the maximum likelihood estimator for ϕ , satisfying $(\partial L_n/\partial \phi^i)(\theta_0, \bar{\phi}_n) = 0$. The existence and consistency of $\bar{\phi}_n$ follows from the classical results for regular estimation problems. Similarly, let $\bar{\theta}_n \equiv \bar{\theta}_n(\phi_0)$ denote the maximum likelihood estimator for θ when $\phi = \phi_0$ is known. The asymptotic properties of $\bar{\theta}_n$ are given by the results in §1. In particular, $\bar{\theta}_n$ exists and is consistent when $\alpha > 1$.

Define $\xi_{n,\beta}$ ($n \geq 1$) to be 1 if $\beta > 1$, $\log n$ if $\beta = 1$ and $n^{1/\beta-1}$ if $0 < \beta < 1$, and write $Y_n <_p r_n$, for random variables $\{Y_n\}$ and positive constants $\{r_n\}$, if

$$\lim_{a \rightarrow \infty} \limsup_{n \rightarrow \infty} \text{pr}(|Y_n| > ar_n) = 0. \quad (3.1)$$

Our main result in this section is the following.

THEOREM 1. *Assume Assumptions 1 and 6–8 are satisfied, and that $\alpha > 1$. Suppose that*

M is strictly positive-definite, where:

- (i) for $\alpha > 2$, M is the $(p+1) \times (p+1)$ matrix with entries $m_{ij}(\phi_0)$ ($i, j = 0, \dots, p$);
- (ii) for $1 < \alpha \leq 2$, M is the $p \times p$ matrix with entries $m_{ij}(\phi_0)$ ($i, j = 1, \dots, p$), the functions m_{ij} being defined in Assumption 6. Then there exists a sequence $(\hat{\theta}_n, \hat{\phi}_n)$ of solutions to the likelihood equations such that

$$\hat{\theta}_n - \theta_0 <_p (n \zeta_{n,\alpha})^{-\frac{1}{\alpha}}, \quad \hat{\phi}_n - \phi_0 <_p n^{-\frac{1}{\alpha}}.$$

Moreover, if $\alpha = 2$ we have

$$\hat{\theta}_n - \bar{\theta}_n <_p n^{-\frac{1}{2}} (\log n)^{-1}, \quad \hat{\phi}_n - \bar{\phi}_n <_p (n \log n)^{-\frac{1}{2}},$$

while if $1 < \alpha < 2$ we have

$$\hat{\theta}_n - \bar{\theta}_n <_p n^{-2/\alpha + \frac{1}{2}}, \quad \hat{\phi}_n - \bar{\phi}_n <_p n^{-1/\alpha}.$$

Theorem 1 is a result about the local behaviour of the likelihood function near the true parameter values. There is no guarantee that the local maximum, whose existence is guaranteed by Theorem 1 when $\alpha > 1$, is unique, and we know already that the local maximum is not a global maximum. The following result goes part of the way to settling the uniqueness of the estimator. It is an analogue for this problem of Wald's (1949) result on the consistency of the maximum likelihood estimator in regular estimation problems.

THEOREM 2. Suppose that, for each n , we have a sample of n independent observations from the density $f(\cdot; \theta_0, \phi_0)$, ordered $X_{n,1} < \dots < X_{n,n}$. For fixed $\delta > 0, \varepsilon > 0, \gamma < \infty$, define the following regions of the parameter space $\mathbb{R} \times \Phi$: U is the set of (θ, ϕ) for which $\theta < \theta_0 - \delta$, $\phi \in \Phi$, and V_n is the set of (θ, ϕ) for which $|\theta - \theta_0| < \delta, |\phi - \phi_0| > \delta$ and either $\alpha(\phi) > 1 + \varepsilon$ or $\theta < X_{n,1} - n^{-\gamma}$.

- (i) Suppose Assumptions 1–6 hold with $\delta^* = \delta$ in Assumption 3, $\delta^1 = \delta$ in Assumptions 4 and 5. Then

$$\lim_{n \rightarrow \infty} \text{pr} \left\{ \sup_U L_n(\theta, \phi) \geq L_n(\theta_0, \phi_0) \right\} = 0.$$

- (ii) Suppose Assumptions 1–6 hold with $\delta^* = 0$ in Assumption 3, $\delta' = \delta$ in Assumptions 4 and 5, and $\alpha > 1$. Then

$$\lim_{n \rightarrow \infty} \text{pr} \left\{ \sup_{V_n} L_n(\theta, \phi) \geq L_n(\theta_0, \phi_0) \right\} = 0.$$

We remark that the main interest in this result lies in the case $\alpha > 1$, when (i) and (ii) both hold. The theorem then shows that the region on which $X_{n,1} - \theta$ is exponentially small and $\alpha(\phi) \leq 1$ is asymptotically the only region where the likelihood function is badly behaved. When $\alpha < 1$, for all our four examples the log likelihood is J-shaped, which shows at once that there can be no consistent maximum likelihood estimator.

Note that we have not settled the question of whether there is a unique local maximum of the likelihood function. The proof of Theorem 1 makes it clear that the Hessian of the log likelihood is negative-definite on a small neighbourhood depending on n around (θ_0, ϕ_0) , and results of Mäkeläinen, Schmidt & Styan (1981) then show that there is a unique local maximum on this neighbourhood. The question of global uniqueness is much harder to resolve.

We start the proofs with several technical lemmas. In Lemmas 1–3, the only assumptions made are Assumption 1 and (2.1). Convergence of random variables is always convergence in distribution unless stated; $\rightarrow_p$ means convergence in probability.

LEMMA 1. Let $X_1, \dots, X_n$ be independent from $f(\cdot; \theta_0, \phi_0)$ and let

$$S_{n,m} = \sum_{k=1}^n (X_k - \theta_0)^{-m} \quad (m > 0).$$

Then $a_n^{-1}(S_{n,m} - b_n) \rightarrow W$ for some a_n, b_n and W having a stable distribution with index $\min(2, \alpha/m)$. We may take $a_n = (n\xi_{n, \frac{1}{2}\alpha/m})^{\frac{1}{\alpha}}$. If $m = \alpha$ then also

$$S_{n,m}/(cn \log n) \rightarrow_p 1.$$

Proof. The random variable $(X_i - \theta_0)^{-m}$ has a distribution in a stable domain of attraction. The conclusion follows from standard results (Feller, 1971, §§ XVII.5, VII.7). $\square$

LEMMA 2. Let $X_1, \dots, X_n$ be as in Lemma 1, ordered by $X_{n,1} < \dots < X_{n,n}$. Define

$$S_{n,m}^* = \sum_{k=2}^n (X_{n,k} - X_{n,1})^{-m} \quad (m > 0).$$

- (i) If $\alpha \geq m$, then $S_{n,m}^*/S_{n,m} \rightarrow_p 1$ as $n \rightarrow \infty$.
- (ii) If $\alpha < m$, then $S_{n,m}^* <_p n^{m/\alpha}$ as $n \rightarrow \infty$.

Proof. Define $S_{n,m}^{**} = S_{n,m} - (X_{n,1} - \theta_0)^{-m}$. It is easily seen that $S_{n,m}^*/S_{n,m} \rightarrow_p 1$, so we concentrate on showing that $S_{n,m}^*/S_{n,m}^{**} \rightarrow_p 1$. Now

$$(X_{n,k} - X_{n,1})^{-m} - (X_{n,k} - \theta_0)^{-m} \leq (X_{n,k} - \theta_0)^{-m} h\{(X_{n,1} - \theta_0)/(X_{n,k} - \theta_0)\},$$

where $h(t) = mt(1-t)^{-m-1}$ for $0 < t < 1$. Note that $h(t)$ is increasing in t and tends to 0 as $t \rightarrow 0$. Therefore

$$S_{n,m}^* - S_{n,m}^{**} \leq \sum_{k=2}^n (X_{n,k} - \theta_0)^{-m} h\{(X_{n,1} - \theta_0)/(X_{n,k} - \theta_0)\}. \quad (3.2)$$

Part (i) follows by splitting the sum in (3.2) into two parts, using the monotonicity of h and $(X_{n,k} - \theta_0)^{-m}/S_{n,m}^{**} \rightarrow_p 0$ for any fixed k . Part (ii) is similar but easier. $\square$

For the results which follow, we need some new notation. Suppose $\{Y_n(\lambda_n), n \geq 1\}$ is a random sequence indexed by $\lambda_n \in \Lambda_n$, and $\{r_n, n \geq 1\}$ is a sequence of positive constants. We shall say that $Y_n(\lambda_n) <_p r_n$ uniformly in Λ_n if the relation (3.1) holds uniformly over $\lambda_n \in \Lambda_n$. Similarly, we say that $Y_n(\lambda_n) \rightarrow_p c$ uniformly in Λ_n if convergence in probability holds uniformly over Λ_n .

Define

$$Y_n^{(i)}(\theta) = \begin{cases} -n^{-1} \sum_{k=1}^n \log(X_{n,k} - \theta) & (i = 1), \\ n^{-1} \sum_{k=1}^n (X_{n,k} - \theta)^{-i+1} & (i = 2, 3). \end{cases}$$

LEMMA 3. Given positive sequences $\{\delta_n\}, \{\delta'_n\}$, the following relations hold uniformly over θ_n satisfying $\theta_n < X_{n,1} - \delta_n, |\theta_0 - \theta_n| < \delta'_n$:

- (i) $|Y_n^{(1)}(\theta_n) - Y_n^{(1)}(\theta_0)| <_p \max\{(n \log 1/\delta_n)^{-1}, n^{-1} \log n, \delta'_n \xi_{n,\alpha}\};$
- (ii) $|Y_n^{(2)}(\theta_n) - Y_n^{(2)}(\theta_0)| <_p \max\{(n\delta_n)^{-1}, n^{1/\alpha-1}, \delta'_n \xi_{n, \frac{1}{2}\alpha}\};$
- (iii) $|Y_n^{(3)}(\theta_n) - Y_n^{(3)}(\theta_0)| <_p \max\{n^{-1} \delta_n^{-2}, n^{2/\alpha-1}, \delta'_n \xi_{n,\alpha/3}\}.$

Proof. We prove only (ii); (i) and (iii) are similar. We may write $Y_n^{(2)}(\theta_n) - Y_n^{(2)}(\theta_0)$ as the sum of three terms: $n^{-1}(X_{n,1} - \theta_n)^{-1}$, $-n^{-1}(X_{n,1} - \theta_0)^{-1}$,

$$n^{-1} \sum_{k=2}^n \{(X_{n,k} - \theta_n)^{-1} - (X_{n,k} - \theta_0)^{-1}\}.$$

The first term is $<_p (n\delta_n)^{-1}$ and the second is $<_p n^{1/\alpha-1}$. For the third term,

$$|(X_{n,k} - \theta_n)^{-1} - (X_{n,k} - \theta_0)^{-1}| \leq |\theta_n - \theta_0| / (X_{n,k} - X_{n,1})^2.$$

But it follows from Lemmas 1 and 2 that

$$n^{-1} \sum_{k=2}^n (X_{n,k} - X_{n,1})^{-2} <_p \xi_{n, \frac{1}{2}\alpha}.$$

Hence this third term is $<_p \delta'_n \xi_{n, \frac{1}{2}\alpha}$. The result follows by putting together these three bounds. $\square$

LEMMA 4. Let δ_n, δ_n^* for $n = 1, 2, \dots$, be any sequences of positive numbers with $\delta_n \rightarrow 0, \delta_n^* \rightarrow \infty$, and let b, b' be any positive finite numbers. Let Assumptions 1 and 6–8 hold.

I (i) Suppose $\alpha > 2$. Then $-(\partial^2/\partial\theta^2) L_n(\theta, \phi) \rightarrow_p m_{00}(\phi_0)$ uniformly over

$$|\theta - \theta_0| < bn^{-1/\alpha}, \quad |\phi - \phi_0| < \delta_n, \quad \theta < X_{n,1} - \delta_n^* n^{-\frac{1}{2}}.$$

I (ii) Suppose $\alpha = 2$. Then $-(\log n)^{-1}(\partial^2/\partial\theta^2) L_n(\theta, \phi) \rightarrow_p c$ uniformly over

$$|\theta - \theta_0| < bn^{-\frac{1}{2}}, \quad |\phi - \phi_0| < \delta_n, \quad \theta < X_{n,1} - \delta_n^* (n \log n)^{-\frac{1}{2}}.$$

I (iii) Suppose $\alpha < 2$. Then $-(\partial^2/\partial\theta^2) L_n(\theta, \phi) <_p n^{2/\alpha-1}$ uniformly over

$$|\theta - \theta_0| < bn^{-1/\alpha}, \quad |\phi - \phi_0| < \delta_n, \quad \theta < X_{n,1} - b'n^{-1/\alpha}.$$

Moreover $-n^{-2/\alpha+1}(\partial^2/\partial\theta^2) L_n(\theta, \phi) > Z_n$ on this range, where $\{Z_n\}$ is a sequence of asymptotically positive random variables in the sense that

$$\lim_{a \downarrow 0} \lim_{n \rightarrow \infty} \text{pr}(Z_n > a) = 1.$$

II (i) Suppose $\alpha > 1$. Then $(\partial^2/\partial\theta \partial\phi^i) L_n(\theta, \phi) \rightarrow_p m_{0i}(\phi_0)$ for $i = 1, \dots, p$, uniformly over

$$|\theta - \theta_0| < \delta_n/\xi_{n, \frac{1}{2}\alpha}, \quad |\phi - \phi_0| < \delta_n, \quad \theta < X_{n,1} - n^{-1} \delta_n^*.$$

II (ii) Suppose $\alpha = 1$. Then $(\partial^2/\partial\theta \partial\phi^i) L_n(\theta, \phi) <_p \log n$, for $i = 1, \dots, p$, uniformly over

$$|\theta - \theta_0| < b(\log n)/n, \quad |\phi - \phi_0| < \delta_n, \quad \theta < X_{n,1} - b'/(n \log n).$$

II (iii) Suppose $\alpha < 1$. Then $(\partial^2/\partial\theta \partial\phi^i) L_n(\theta, \phi) <_p n^{1/\alpha-1}$ uniformly over

$$|\theta - \theta_0| < bn^{-1/\alpha}, \quad |\phi - \phi_0| < \delta_n, \quad \theta \leq X_{n,1} - b'n^{-1/\alpha}.$$

III For $\alpha > 0$, $(\partial^2/\partial\phi^i \partial\phi^j) L_n(\theta, \phi) \rightarrow_p m_{ij}(\phi_0)$ for $i, j = 1, \dots, p$, uniformly over $|\theta - \theta_0| < \delta_n/\xi_{n, \alpha}, |\phi - \phi_0| < \delta_n, \theta < X_{n,1} - n^{-b}$. If the upper bound on $|\theta - \theta_0|$ is relaxed to $b'/\xi_{n, \alpha}$, we still have $(\partial^2/\partial\phi^i \partial\phi^j) L_n(\theta, \phi) <_p 1$ uniformly.

Proof. I. We write

$$-\left(\frac{\partial^2}{\partial\theta^2}\right) L_n(\theta, \phi) = \{\alpha(\phi) - 1\} Y_n^{(3)}(\theta) - n^{-1} \sum_{k=1}^n \left(\frac{\partial^2}{\partial\theta^2}\right) \log g(X_{n,k} - \theta; \phi). \quad (3.3)$$

- (i) The result is true for $\theta = \theta_0, \phi = \phi_0$ by Assumption 6. Therefore it suffices to prove $(\partial^2/\partial\theta^2)\{L_n(\theta, \phi) - L_n(\theta_0, \phi_0)\} \rightarrow_p 0$ uniformly. By (3.3) and Assumption 7

$$\left| \left(\frac{\partial^2}{\partial\theta^2} \right) \{L_n(\theta, \phi) - L_n(\theta_0, \phi_0)\} \right| \leq |\{\alpha(\phi) - 1\}Y_n^{(3)}(\theta) - \{\alpha(\phi_0) - 1\}Y_n^{(3)}(\theta_0)| \\ + n^{-1} \left| \sum_{k=1}^n \left(\frac{\partial^2}{\partial x^2} \right) \{\log g(X_{n,k} - \theta; \phi) - \log g(X_{n,k} - \theta_0; \phi_0)\} \right|$$

and the second term tends to zero in probability. By Lemma 3(iii) and the continuity of $\alpha(\cdot)$, the first term also tends to zero in probability, uniformly as required.

- (ii) For arbitrarily small ε, δ , the second term in (3.3) is bounded by

$$\varepsilon Y_n^{(3)}(\theta) + n^{-1} \sum_{k=1}^n h_{\varepsilon, \delta}(X_{n,k} - \theta_0, \phi_0)$$

uniformly over $|\theta - \theta_0| < \delta, |\phi - \phi_0| < \delta$, where the latter term is $O_p(1)$, by Assumption 8. The result then follows from Lemmas 1 and 3(iii).

- (iii) The first part again follows by a combination of Assumption 8 and Lemma 3(iii), together with Lemma 1. For the second part, note that $-(\partial^2 L_n/\partial\theta^2)$ is bounded below, up to an error of at most $O_p(1)$, by

$$(\alpha - 1)n^{-1}(X_{n,1} - \theta)^{-2} \geq (\alpha - 1)n^{-1}(X_{n,1} - \theta_0 + b'n^{-1/\alpha})^{-2}.$$

With $W_n = n^{1/\alpha}(X_{n,1} - \theta_0)$, this becomes $(\alpha - 1)n^{2/\alpha-1}(W_n - b')^{-2}$. But W_n converges to a nondegenerate Weibull limit by extreme value theory. Hence the stated conditions are satisfied by $Z_n = (\alpha - 1)(W_n + b')^{-2}$. $\square$

We shall not give the proofs of the remaining parts of Lemma 4. They follow by arguments similar to those already given, using Assumptions 1, 6 and 7 and the relevant parts of Lemmas 1 and 3.

LEMMA 5. *Let h be a continuously differentiable real-valued function of $p+1$ real variables and let H denote the gradient vector of h . Suppose that the scalar product of x and $H(x)$ is negative whenever $|x| = 1$. Then h has a local maximum, at which $H = 0$, for some x with $|x| < 1$.*

The proof is omitted.

Proof of Theorem 1. The case $\alpha > 2$ is straightforward, so that we do this first. Let $\{\delta_n\}$ be any sequence such that $n^{1/\alpha}\delta_n \rightarrow 0, n^{\frac{1}{2}}\delta_n \rightarrow \infty$ and define for $t \in R, y \in \mathbb{R}^p$,

$$f_n(t, y) = \delta_n^{-2} L_n(\theta_0 + \delta_n t, \phi_0 + \delta_n y).$$

By expanding $\partial f_n/\partial t$ and $\partial f_n/\partial y^j$ as far as the second term and using the results of Lemma 4, we deduce that

$$\partial f_n(t, y)/\partial t = -tm_{00}(\phi_0) - \sum_i y^i m_{0i}(\phi_0) + \varepsilon_{n,0}(t, y),$$

$$\partial f_n(t, y)/\partial y^j = -tm_{0j}(\phi_0) - \sum_i y^i m_{ij}(\phi_0) + \varepsilon_{n,j}(t, y),$$

where $\varepsilon_{n,j}(t, y) \rightarrow_p 0$ uniformly over $t^2 + |y|^2 \leq 1$ say, for $j = 0, \dots, p$.

Let $t^2 + |y|^2 = 1$. We have

$$t\partial f_n/\partial t + \sum y^j \partial f_n/\partial y^j = -t^2 m_{00} - 2t \sum y^j m_{0j} - \sum \sum y^i y^j m_{ij} + t\varepsilon_{n,0} + \sum y^j \varepsilon_{n,j}$$

which is as $n \rightarrow \infty$ eventually strictly negative by the assumed positive-definiteness of M . Hence Lemma 5 shows that f_n has a local maximum in the range $t^2 + |y|^2 < 1$, with probability tending to 1 as $n \rightarrow \infty$. Since the sequence $n^{\frac{1}{2}} \delta_n$ may be made to increase arbitrarily slowly, this proves the result for the case $\alpha > 2$.

Now let $\alpha \leq 2$. The argument proceeds by showing that $\hat{\theta}_n, \hat{\phi}_n$ are respectively near $\bar{\theta}_n, \bar{\phi}_n$. For $1 \leq i \leq p$,

$$\begin{aligned} \frac{\partial L_n}{\partial \phi^i}(\theta, \phi) &= \frac{\partial L_n}{\partial \phi^i}(\theta, \phi) - \frac{\partial L_n}{\partial \phi^i}(\theta_0, \bar{\phi}_n) \\ &= (\theta - \theta_0) \frac{\partial^2 L_n}{\partial \theta \partial \phi^i}(\theta^*, \phi^*) + \sum_j (\phi^j - \bar{\phi}_n^j) \frac{\partial L_n}{\partial \phi^i \partial \phi^j}(\theta^*, \phi^*), \end{aligned}$$

where $(\theta^*, \phi^*) = \lambda(\theta, \phi) + (1 - \lambda)(\theta_0, \bar{\phi}_n)$ for some λ between 0 and 1. Thus we may write

$$\partial L_n(\theta, \phi) / \partial \phi^i = -(\theta_n - \theta_0) m_{0i} - \sum_j (\phi^j - \bar{\phi}_n^j) m_{ij} + e_n(\theta, \phi). \quad (3.4)$$

Similarly,

$$\begin{aligned} \frac{\partial L_n}{\partial \theta}(\theta, \phi) &= \left\{ \frac{\partial L_n}{\partial \theta}(\theta, \bar{\phi}_n) - \frac{\partial L_n}{\partial \theta}(\bar{\theta}_n, \bar{\phi}_n) \right\} + \left\{ \frac{\partial L_n}{\partial \theta}(\theta, \phi) - \frac{\partial L_n}{\partial \theta}(\theta, \bar{\phi}_n) \right\} \\ &\quad + \left\{ \frac{\partial L_n}{\partial \theta}(\bar{\theta}_n, \bar{\phi}_n) - \frac{\partial L_n}{\partial \theta}(\bar{\theta}_n, \phi_0) \right\} \\ &= \left\{ \frac{\partial L_n}{\partial \theta}(\theta, \bar{\phi}_n) - \frac{\partial L_n}{\partial \theta}(\bar{\theta}_n, \bar{\phi}_n) \right\} - \sum_i (\bar{\phi}_n^i - \phi_0^i) m_{0i} + e_n^*(\theta, \phi). \end{aligned} \quad (3.5)$$

Let $\bar{\theta}_n^*$ satisfy

$$\frac{\partial L_n}{\partial \theta}(\bar{\theta}_n^*, \bar{\phi}_n) - \frac{\partial L_n}{\partial \theta}(\bar{\theta}_n, \bar{\phi}_n) = \sum_i (\bar{\phi}_n^i - \phi_0^i) m_{0i}. \quad (3.6)$$

For the moment, we assume $\bar{\theta}_n^*$ exists. Comparing (3.6) with (3.5), we see

$$\frac{\partial L_n}{\partial \theta}(\theta, \phi) = \frac{\partial L_n}{\partial \theta}(\theta, \bar{\phi}_n) - \frac{\partial L_n}{\partial \theta}(\bar{\theta}_n^*, \bar{\phi}_n) + e_n^*(\theta, \phi) = (\theta - \bar{\theta}_n^*) \frac{\partial^2 L_n}{\partial \theta^2}(\theta^{**}, \bar{\phi}_n) + e_n^*(\theta, \phi) \quad (3.7)$$

for some θ^{**} between θ and $\bar{\theta}_n^*$. If we also define $\bar{\phi}_n^*$ to satisfy

$$0 = -(\bar{\theta}_n - \theta_0) m_{0i} - \sum_j (\bar{\phi}_n^{*j} - \bar{\phi}_n^j) m_{ij},$$

(3.4) becomes

$$\frac{\partial L_n}{\partial \phi^i}(\theta, \phi) = -\sum_j (\phi^j - \bar{\phi}_n^{*j}) m_{ij} + e_n(\theta, \phi). \quad (3.8)$$

Note that $\bar{\phi}_n^*$ must exist, since M is invertible.

To show that $\bar{\theta}_n^*$ exists, with probability tending to 1, we may assume $\alpha(\bar{\phi}_n) > 1$ for large n , so that $(\partial L_n / \partial \theta)(\theta, \bar{\phi}_n) \rightarrow -\infty$ as $\theta \uparrow X_{n,1}$. Since $\partial L_n / \partial \theta$ is continuous, it suffices to show that there exists a θ with

$$\frac{\partial L_n}{\partial \theta}(\theta, \bar{\phi}_n) - \frac{\partial L_n}{\partial \theta}(\bar{\theta}_n, \bar{\phi}_n) > \sum_i (\bar{\phi}_n^i - \phi_0^i) m_{0i}.$$

The right-hand side is $O_p(n^{-\frac{1}{2}})$ while the left is of the form $(\theta - \bar{\theta}_n)(\partial^2 L_n / \partial \theta^2)(\theta_1, \bar{\phi}_n)$ for

some θ_1 between θ and $\bar{\theta}_n$. For $\alpha = 2$, we have $-\log n (\partial^2/\partial\theta^2) L_n(\theta_1, \bar{\phi}_n) \rightarrow_p c$ uniformly on $|\theta_1 - \theta_0| < n^{-\frac{1}{2}}$ by Lemma 4I(ii), hence $\bar{\theta}_n^*$ exists and satisfies $\bar{\theta}_n^* - \bar{\theta}_n <_p n^{-\frac{1}{2}}(\log n)^{-1}$ as $n \rightarrow \infty$. Similarly, for $1 < \alpha < 2$ we apply Lemma 4I(iii) to deduce that $\bar{\theta}_n^*$ exists and that $\bar{\theta}_n^* - \bar{\theta}_n <_p n^{-2/\alpha + \frac{1}{2}}$.

Now consider e_n and e_n^* . For $\alpha = 2$, we have $\bar{\theta}_n - \theta_0 <_p (n \log n)^{-\frac{1}{2}}$ and it follows that $(n \log n)^{\frac{1}{2}} e_n(\theta_n, \phi_n) \rightarrow_p 0$ along any sequence (θ_n, ϕ_n) such that

$$(n \log n)^{\frac{1}{2}} (\theta_n - \bar{\theta}_n) \rightarrow_p 0, \quad \phi_n - \bar{\phi}_n <_p (n \log n)^{-\frac{1}{2}}.$$

Under the same conditions, we have also $n^{\frac{1}{2}} e_n^*(\theta_n, \phi_n) \rightarrow 0$. For $1 < \alpha < 2$,

$$\bar{\theta}_n - \theta_0 <_p n^{-1/\alpha}$$

and it follows that $n^{1/\alpha} e_n(\theta_n, \phi_n) \rightarrow_p 0$ and $n^{\frac{1}{2}} e_n^*(\theta_n, \phi_n) \rightarrow_p 0$ along any sequence (θ_n, ϕ_n) such that $n^{1/\alpha}(\theta_n - \bar{\theta}_n) \rightarrow_p 0$ and $\phi_n - \bar{\phi}_n <_p n^{-1/\alpha}$.

We are ready for the final step. Suppose $\alpha = 2$. For $t \in \mathbb{R}, y \in \mathbb{R}^p$, define

$$f_n(t, y) = n \log n L_n\{\bar{\theta}_n^* + tn^{-\frac{1}{2}}(\log n)^{-1}, \bar{\phi}_n^* + y(n \log n)^{-\frac{1}{2}}\}.$$

Then

$$\begin{aligned} \frac{\partial f_n}{\partial t} &= n^{\frac{1}{2}} \frac{\partial L_n}{\partial \theta} \{\bar{\theta}_n^* + tn^{-\frac{1}{2}}(\log n)^{-1}, \bar{\phi}_n^* + y\{n \log n\}^{-\frac{1}{2}}\} \\ &= t(\log n)^{-1} \frac{\partial^2 L_n}{\partial \theta^2}(\theta^{**}, \bar{\phi}_n) + n^{\frac{1}{2}} e_n^*\{\bar{\theta}_n^* + tn^{-\frac{1}{2}}(\log n)^{-1}, \bar{\phi}_n^* + y(n \log n)^{-\frac{1}{2}}\} \end{aligned}$$

by (3·7). But the second term tends to zero in probability, and the first to $-tc$. Similarly, $\partial f_n/\partial y^i \rightarrow_p -\sum_j y^j m_{ij}$ uniformly over $t^2 + |y|^2 \leq 1$, say, where we use (3·8) and the previously stated result about e_n . Note that

$$\bar{\theta}_n^* - \bar{\theta}_n <_p n^{-\frac{1}{2}}(\log n)^{-1} = o\{(n \log n)^{-\frac{1}{2}}\}, \quad \bar{\phi}_n^* - \bar{\phi}_n <_p (n \log n)^{-\frac{1}{2}},$$

so the required conditions are satisfied. Then

$$t \partial f_n/\partial t + \sum y^i \partial f_n/\partial y^i \rightarrow_p -tc - \sum \sum y^i y^j m_{ij} < 0,$$

so that, applying Lemma 5, the probability that f_n has a local maximum within the ball $t^2 + |y|^2 \leq \delta^2$, for any fixed $\delta > 0$, tends to 1 as $n \rightarrow \infty$. Hence L_n has a local maximum at $(\hat{\theta}_n, \hat{\phi}_n)$ satisfying

$$\hat{\theta}_n - \bar{\theta}_n <_p n^{-\frac{1}{2}}(\log n)^{-1}, \quad \hat{\phi}_n - \bar{\phi}_n <_p (n \log n)^{-\frac{1}{2}} \quad (3\cdot9)$$

with probability tending to 1.

The case $1 < \alpha < 2$ is similar to the case $\alpha = 2$. Now define

$$f_n(t, y) = n^{2/\alpha} L_n(\bar{\theta}_n^* + tn^{-2/\alpha + \frac{1}{2}}, \bar{\phi}_n^* + yn^{-1/\alpha}).$$

Expanding as far as the second term, using (3·7), (3·8) and Lemma 4I(iii), we may show that $t \partial f_n/\partial n + \sum y^i \partial f_n/\partial y^i$ is strictly negative over $t^2 + |y|^2 = \delta^2$, with probability tending to 1 as $n \rightarrow \infty$. Again Lemma 5 may be applied, and we conclude that f_n has a local maximum satisfying $t^2 + |y|^2 < \delta^2$, for any $\delta > 0$, with probability tending to 1. Hence L_n has a local maximum at $(\hat{\theta}_n, \hat{\phi}_n)$ satisfying

$$\hat{\theta}_n - \bar{\theta}_n <_p n^{-2/\alpha + \frac{1}{2}}, \quad \hat{\phi}_n - \bar{\phi}_n <_p n^{-1/\alpha},$$

with probability tending to 1 as $n \rightarrow \infty$.

The proof of Theorem 1 is complete. □

Proof of Theorem 2. (i) This follows Wald's classical proof very closely; see also Walker (1969) where similar results are needed as a preliminary to establishing asymptotic normality of the Bayes estimator in a regular case. There are three steps.

First, for fixed $(\theta_1, \phi_1) \in U$ there exists $b(\theta_1, \phi_1) > 0$ such that

$$\lim_{n \rightarrow \infty} \text{pr} \{L_n(\theta_1, \phi_1) - L_n(\theta_0, \phi_0) < -b(\theta_1, \phi_1)\} = 1.$$

This uses Assumption 2 and Jensen's inequality.

Secondly, for fixed $(\theta_1, \phi_1) \in U$, η between 0 and $|\theta_1 - \theta_0| - \delta$, we have

$$\lim_{n \rightarrow \infty} \text{pr} \{\sup L_n(\theta, \phi) < L_n(\theta_0, \phi_0) - b(\theta_1, \phi_1)/4\} = 1,$$

where the supremum is over all (θ, ϕ) such that $|\theta - \theta_1| < \eta, |\phi - \phi_1| < \eta$. This is proved by bounding $L_n(\theta, \phi) - L_n(\theta_1, \phi_1)$, using Assumptions 1 and 3 the first step above.

Thirdly, let K_m be a compact subset of $\mathbb{R} \times \Phi$, as in Assumption 4. Extending the result of the second step above, we have

$$\lim_{n \rightarrow \infty} \text{pr} \left\{ \sup_{K_m \cap U} L_n(\theta, \phi) < L_n(\theta_0, \phi_0) - \eta_m \right\} = 1$$

for some $\eta_m > 0$. This is because K_m can be covered by a finite number of open neighbourhoods of points (θ_1, ϕ_1) . But now Assumption 4 allows us to drop the restriction to K_m , provided m is sufficiently large.

(ii) First we note that, if $\theta = \theta_0$ is assumed known then, given $\delta > 0$, there exists $\xi > 0$ such that

$$\lim_{n \rightarrow \infty} \text{pr} \left\{ \sup_{\phi: |\phi - \phi_0| > \delta} L_n(\theta_0, \phi) < L_n(\theta_0, \phi_0) - \xi \right\} = 1. \quad (3.10)$$

This may be proved by imitating the arguments used to prove (i), using Assumptions 2 and 3 with $\varepsilon = \varepsilon_1 = 0$, and Assumption 5.

Let K'_m be a compact subset of Φ , as in Assumption 5. For

$$(\theta, \phi_1) \in V_n, \quad \phi \in K'_m, \quad \phi_1 \in K'_m,$$

we have

$$\begin{aligned} L_n(\theta, \phi) - L_n(\theta_0, \phi_1) &= n^{-1} \{ \alpha(\phi) - 1 \} \sum_k \{ \log(X_{n,k} - \theta) - \log(X_{n,k} - \theta_0) \} \\ &\quad + n^{-1} \{ \alpha(\phi) - \alpha(\phi_1) \} \sum_k \log(X_{n,k} - \theta_0) \\ &\quad + n^{-1} \sum_k \{ \log g(X_{n,k} - \theta; \phi) - \log g(X_{n,k} - \theta_0; \phi_1) \}. \end{aligned} \quad (3.11)$$

But if $|\phi - \phi_1| < \eta, \delta \leq \eta$,

$$n^{-1} \sum_k \{ \log g(X_{n,k} - \theta; \phi) - \log g(X_{n,k} - \theta_0; \phi_1) \} < n^{-1} \sum_k H_\eta(X_{n,k} - \theta_0; \phi_1)$$

and by Assumption 3 we may choose η and hence δ sufficiently small that

$$\lim_{n \rightarrow \infty} \text{pr} \{ n^{-1} \sum_k H_\eta(X_{n,k} - \theta_0; \phi_1) > \xi/4 \} = 1. \quad (3.12)$$

Note that it is essential that Assumption 3 holds with $\delta^* = 0$ for this step.

Define an event $\mathcal{E}_{n,\varepsilon}(\theta, \phi)$ to hold if and only if

$$n^{-1} \{ \alpha(\phi) - 1 \} \sum_k \{ \log(X_{n,k} - \theta) - \log(X_{n,k} - \theta_0) \} < \varepsilon \{ \alpha(\phi) + 1 \}.$$

We claim that, for any $\varepsilon > 0$, it is possible to choose δ sufficiently small so that

$$\lim_{n \rightarrow \infty} \text{pr} \{ \mathcal{E}_{n,\varepsilon}(\theta, \phi) \text{ for all } (\theta, \phi) \in V_n \} = 1. \quad (3.13)$$

To prove this, fix $(\theta, \phi) \in V_n$ and consider two cases: (a) $\alpha(\phi) > 1$, (b) $X_{n,1} < \theta - n^{-\gamma}$. We use the inequality

$$|\log(x - \theta_1) - \log(x - \theta_0)| < |\theta_1 - \theta_0| / \{x - \max(\theta_1, \theta_0)\}$$

valid whenever $x > \max(\theta_1, \theta_0)$. In case (a),

$$n^{-1} \{\alpha(\phi) - 1\} \sum_k \{\log(X_{n,k} - \theta) - \log(X_{n,k} - \theta_0)\}$$

is negative when $\theta > \theta_0$, and is bounded by $n^{-1} \delta \{\alpha(\phi) - 1\} \sum_k (X_{n,k} - \theta_0)^{-1}$ when $\theta < \theta_0$. Since $E\{(X - \theta_0)^{-1}\} < \infty$, we may choose δ sufficiently small, independently of ϕ , so that

$$\lim_{n \rightarrow \infty} \text{pr}[n^{-1} \delta \{\alpha(\phi) - 1\} \sum_k (X_{n,k} - \theta_0)^{-1} > \varepsilon \{\alpha(\phi) + 1\}] = 0.$$

In case (b),

$$n^{-1} \sum_k |\log(X_{n,k} - \theta) - \log(X_{n,k} - \theta_0)| \leq n^{-1} \gamma \log n + \delta n^{-1} \sum_{k=2}^n (X_{n,k} - X_{n,1})^{-1}$$

and the last term converges in probability by Lemma 2. Putting the results for (a) and (b) together, we have (3·13).

We also have $E\{|\log(X_{n,k} - \theta_0)|\} < \infty$ and may make $|\alpha(\phi) - \alpha(\phi_1)|$ arbitrarily small by choosing η sufficiently small. Combining this observation with (3·10), (3·12) and (3·13), choosing ε in (3·13) so that $\varepsilon \{\alpha(\phi) + 1\} < \xi/4$ on the range $|\phi - \phi_1| < \eta$, we have

$$\lim_{n \rightarrow \infty} \text{pr}\{\sup L_n(\theta, \phi) < L_n(\theta_0, \phi_0) - \xi/4\} = 1, \quad (3·14)$$

where the supremum is now taken over all (θ, ϕ) such that

$$(\theta, \phi_1) \in V_n, \quad \phi \in K'_m, \quad \phi_1 \in K'_m, \quad |\phi - \phi_1| < \eta,$$

for fixed ϕ_1 . This result may immediately be extended to any finite set of values of ϕ_1 and hence by compactness to the whole of K'_m . Thus (3·14) holds if the supremum is taken over all (θ, ϕ) such that $(\theta, \phi) \in V_n$ and $\phi \in K'_m$.

Now consider the case $(\theta, \phi) \in V_n, \phi \notin K'_m$. Taking $\varepsilon = \eta'$ in (3·13), $\phi_1 = \phi_0$ in (3·11), we have with probability tending to one that

$$L_n(\theta, \phi) - L_n(\theta_0, \phi_0) \leq n^{-1} \sum_k H'_m(X_{n,k} - \theta_0; \phi_0),$$

and the result now follows from Assumption 5. □

4. ASYMPTOTIC DISTRIBUTIONS

We are now in a position to state our main results about the asymptotic distributions of $\hat{\theta}_n$ and $\hat{\phi}_n$.

THEOREM 3. *Under the assumptions of Theorem 1 let $(\hat{\theta}_n, \hat{\phi}_n)$ denote a sequence of maximum likelihood estimators satisfying the conclusions of Theorem 1.*

- (i) *If $\alpha > 2$ then $n^{\frac{1}{2}}(\hat{\theta}_n - \theta_0, \hat{\phi}_n - \phi_0)$ converges in distribution to a normal random vector with mean 0 and covariance matrix M^{-1} , where M is as in Theorem 1(i).*
- (ii) *If $\alpha = 2$ then $\{(nc \log n)^{\frac{1}{2}}(\hat{\theta}_n - \theta_0), n^{\frac{1}{2}}(\hat{\phi}_n - \phi_0)\}$ converges in distribution to a normal random vector with covariance matrix of form*

$$\begin{bmatrix} 1 & 0 \\ 0 & M^{-1} \end{bmatrix},$$

where M is as in Theorem 1(ii).

- (iii) If $1 < \alpha < 2$ then $\{(nc)^{1/\alpha}(\hat{\theta}_n - \theta_0), n^{1/2}(\hat{\phi}_n - \phi_0)\} \rightarrow (Y, Z)$ in distribution, where $Y \in \mathbb{R}$ and $Z \in \mathbb{R}^p$ are independent, Z has a normal distribution with mean 0 and covariance matrix M^{-1} , where M is as in (ii), and the distribution function of Y is H , where H is defined in the statement of Theorem 2.4 of Woodroffe (1974).

Note that Woodroffe's definition of H is long-winded, but because we do not know any way to simplify it, we refer to Woodroffe's paper for the definition.

We also have the following Corollary.

COROLLARY. If $\alpha = 2$ then

$$(\bar{\theta}_n - \theta_0) \{-n \partial^2 L_n(\bar{\theta}_n, \phi_0) / \partial \theta^2\}^{1/2} \rightarrow Z, \quad (4.1)$$

$$(\hat{\theta}_n - \theta_0) \{-n \partial^2 L_n(\hat{\theta}_n, \hat{\phi}_n) / \partial \theta^2\}^{1/2} \rightarrow Z, \quad (4.2)$$

where convergence is in distribution and, in each case, Z is standard normal.

The corollary shows that the variance of the estimators may be estimated asymptotically by means of the observed information, in a case where the expected information does not exist. In regular estimation problems, Efron & Hinkley (1978) argued that the observed information is superior to the expected information as an estimator of variance, but their argument depends on second-order approximations and conditional arguments. It may therefore be of some interest that we have an example in which the superiority of observed information is very easily demonstrated. Our argument, however, applies only to the specific case $\alpha = 2$ and therefore is of only slight practical significance.

Proofs. We require two preliminary lemmas and a remark. Suppose $\{(X_k, Y_k), k \geq 1\}$ is a sequence of independent identically distributed random variables and let (S_n, T_n) denote the sum of (X_k, Y_k) ($k = 1, \dots, n$).

LEMMA 6. Suppose $E(X_1) = E(Y_1) = 0$, $E(X_1^2) = 1$, $E(Y_1^2) = +\infty$ and $S_n/\sqrt{n} \rightarrow Z_1$, $T_n/b_n \rightarrow Z_2$ in distribution, where Z_1 and Z_2 are each standard normal. Then

$$(S_n/\sqrt{n}, T_n/b_n) \rightarrow (Z_1, Z_2)$$

in distribution, with Z_1, Z_2 independent.

Proof. By Theorem 3.1 of an Erasmus University, Rotterdam, technical report by L. de Haan, E. Omey and S. I. Resnick, it suffices to show that the conditional expectations satisfy

$$\frac{E(X_1 Y_1; |Y_1| \leq b_n)}{\{E(Y_1^2; |Y_1| \leq b_n)\}^{1/2}} \rightarrow 0.$$

By splitting into $|X_1| \leq M$ and $|X_1| > M$ and using the Cauchy-Schwarz inequality, this is less than

$$ME(|Y_1|) / \{E(Y_1^2; |Y_1| \leq b_n)\}^{1/2} + \{E(X_1^2; |X_1| > M)\}^{1/2},$$

each term of which may be made arbitrarily small by taking first M and then n sufficiently large. $\square$

LEMMA 7. Suppose $X_1, \dots, X_n$ are independent random variables from a distribution whose density f satisfies $f(x) = 0$ for $x < 0$, $f(x) \sim \alpha c x^{\alpha-1}$ as $x \downarrow 0$. Let

$$M_n = \min(X_1, \dots, X_n), \quad W_n = g(X_1) + \dots + g(X_n),$$

where the function g satisfies $\int g(x) f(x) dx = 0$, $\int g^2(x) f(x) dx = 1$. Let $a_n = (nc)^{-1/\alpha}$. Then $(M_n/a_n, W_n/\sqrt{n})$ converges weakly to (Y, Z) , where Y and Z are independent Weibull and standard normal random variables.

Proof. The case where g is the identity or a linear function is dealt with by Chow & Teugels (1978), and our method closely follows theirs. It suffices to show that

$$E\{\exp(itW_n/\sqrt{n}); M_n > a_n u\} \rightarrow \exp(-\tfrac{1}{2}t^2 - u^\alpha).$$

The left-hand side may be written

$$\left(1 + \int_0^\infty [\exp\{itg(x)/\sqrt{n}\} - 1] f(x) dx - \int_0^{anu} \exp\{itg(x)/\sqrt{n}\} f(x) dx\right)^n.$$

But it follows from the standard proof of the central limit theorem that

$$n \int_0^\infty \{\exp(itg(x)/\sqrt{n}) - 1\} f(x) dx \rightarrow -\tfrac{1}{2}t^2$$

and from the asymptotic form of f that

$$n \int_0^{anu} f(x) dx \rightarrow u^\alpha.$$

The result therefore follows by calculating that

$$\begin{aligned} \left| n \int_0^{anu} [\exp\{itg(x)/\sqrt{n}\} - 1] f(x) dx \right| &< \sqrt{n} \int_0^{anu} |tg(x)| f(x) dx \\ &< \left[\left\{ \int_0^{anu} t^2 g^2(x) f(x) dx \right\} \left\{ n \int_0^{anu} f(x) dx \right\} \right]^{\frac{1}{2}} \\ &\rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$. □

Remark 1. Lemma 6 extends the result of Resnick & Greenwood (1979), Theorem 3, that if $S_n/a_n \rightarrow Z_1$, $T_n/b_n \rightarrow Z_2$ with Z_1 normal, Z_2 stable with index less than 2, then $(S_n/a_n, T_n/b_n) \rightarrow (Z_1, Z_2)$, with Z_1, Z_2 necessarily independent. The key point in the proof is the observation that the limit (Z_1, Z_2) must be infinitely divisible and therefore the sum of independent Gaussian and Lévy components. Our remark is that the same result holds if (Z_1, Z_2) arises as the limit of renormalized row sums of a triangular array subject to the usual asymptotic negligibility condition. That is, if Z_1 is normal and Z_2 has an infinitely divisible distribution without a Gaussian component, then Z_1 and Z_2 are independent.

Proof of Theorem 3. (i) Theorem 1 shows the existence of $(\hat{\theta}_n, \hat{\phi}_n)$ with $\hat{\theta}_n - \theta_0 <_{\mathbf{p}} n^{-\frac{1}{2}}$, $\hat{\phi}_n - \phi_0 <_{\mathbf{p}} n^{-\frac{1}{2}}$, and the proof of Theorem 1 shows that the second derivatives of L_n are asymptotically constant in this region. The result therefore follows by standard arguments.

(ii) Since $(n \log n)^{\frac{1}{2}}(\hat{\theta}_n - \bar{\theta}_n) \rightarrow_{\mathbf{p}} 0$, $n^{\frac{1}{2}}(\hat{\phi}_n - \bar{\phi}_n) \rightarrow_{\mathbf{p}} 0$, it suffices to prove the result with $\bar{\theta}_n, \bar{\phi}_n$ in place of $\hat{\theta}_n, \hat{\phi}_n$. For $\bar{\theta}_n$ alone, the asymptotic distribution is given by Woodroffe (1972). For $\bar{\phi}_n$ alone, the asymptotic distribution is given by the classical results for

regular estimation problems. Therefore the only thing to show is the asymptotic independence of $\bar{\theta}_n$ and $\bar{\phi}_n$.

Now $(nc \log n)^{\frac{1}{2}}(\bar{\theta}_n - \theta_0)$ may be written as

$$(nc \log n)^{\frac{1}{2}} \partial L_n(\theta_0, \phi_0) / \partial \theta \{ -\partial^2 L_n(\theta_n^*, \phi_0) / \partial \theta^2 \}^{-1},$$

where θ_n^* satisfies $|\theta_n^* - \theta_0| <_{\mathbf{p}} (n \log n)^{-\frac{1}{2}}$. By Lemma 4I(ii),

$$-(c \log n)^{-1} \frac{\partial^2 L_n}{\partial \theta^2}(\theta_n^*, \phi_0) \rightarrow_{\mathbf{p}} 1,$$

so we have

$$(nc \log n)^{\frac{1}{2}}(\bar{\theta}_n - \theta_0) = \left(\frac{n}{c \log n} \right)^{\frac{1}{2}} \frac{\partial L_n}{\partial \theta}(\theta_0, \phi_0) V_n,$$

where $V_n \rightarrow_{\mathbf{p}} 1$ and hence plays no role in determining the limit. Now it follows by Lemma 1 and Assumption 9 (Woodroffe, 1972) that $\partial L_n(\theta_0, \phi_0) / \partial \theta$ has infinite variance but that $\{n / (c \log n)\}^{\frac{1}{2}} \partial L_n(\theta_0, \phi_0) / \partial \theta$ converges to a standard normal variable Z_0 say. Similar arguments applied to $(\bar{\phi}_n - \phi_0)$, together with the Cramér–Wold device and Lemma 6, allow us to assert that Z_0 is independent of the asymptotic distribution of $n^{\frac{1}{2}}(\bar{\phi}_n - \phi_0)$, as required.

(iii) Since $n^{1/\alpha}(\hat{\theta}_n - \bar{\theta}_n) \rightarrow_{\mathbf{p}} 0$, $n^{\frac{1}{2}}(\hat{\phi}_n - \bar{\phi}_n) \rightarrow_{\mathbf{p}} 0$, it again suffices to prove the result with $\bar{\theta}_n, \bar{\phi}_n$ in place of $\hat{\theta}_n, \hat{\phi}_n$, and hence the only thing to show is the independence of the asymptotic distributions of $\bar{\theta}_n$ and $\bar{\phi}_n$. Our proof will make use of Remark 1 as well as Lemma 7.

Let $t > 0, y \in \mathbb{R}$ and consider

$$\text{pr} \{ (cn)^{1/\alpha}(\bar{\theta}_n - \theta_0) \leq -t, n^{\frac{1}{2}} \Sigma_j a^j (\bar{\phi}_n^j - \phi_0^j) \leq y \} \quad (4.3)$$

for fixed $a^1, \dots, a^p$. In the notation of Woodroffe (1974) this is the same as

$$\text{pr} \{ Z_{nt} \geq 0, n^{\frac{1}{2}} \Sigma_j a^j (\bar{\phi}_n^j - \phi_0^j) \leq y \},$$

where Z_{nt} ($n \geq 1$) is a sequence of renormalized row sums of a triangular array, converging to an infinitely divisible limit. The limit is given in Woodroffe's Theorem 2.2, and does not contain a Gaussian component. Therefore the limiting probability in (4.3) is the same as that of

$$\text{pr} \{ (cn)^{1/\alpha}(\bar{\theta}_n - \theta_0) \leq -t \} \text{pr} \{ n^{\frac{1}{2}} \Sigma_j a^j (\bar{\phi}_n^j - \phi_0^j) \leq y \}.$$

Similar arguments may be applied to the limits of

$$\begin{aligned} & \text{pr} \{ (cn)^{1/\alpha}(\bar{\theta}_n - \theta_0) \leq 0, n^{\frac{1}{2}} \Sigma_j a^j (\bar{\phi}_n^j - \phi_0^j) \leq y \}, \\ & \text{pr} \{ (cn)^{1/\alpha}(\bar{\theta}_n - \theta_0) > t, n^{\frac{1}{2}} \Sigma_j a^j (\bar{\phi}_n^j - \phi_0^j) \leq y \mid X_{n,1} > \theta_0 + t(cn)^{-1/\alpha} \} \end{aligned}$$

for $t > 0$, using Theorems 2.1 and 2.3 of Woodroffe's paper. The last equation implies that

$$\begin{aligned} & \lim \text{pr} \{ (cn)^{1/\alpha}(\bar{\theta}_n - \theta_0) > t, n^{\frac{1}{2}} \Sigma_j a^j (\bar{\phi}_n^j - \phi_0^j) \leq y \} \\ &= \lim \text{pr} \{ (cn)^{1/\alpha}(\bar{\theta}_n - \theta_0) > t, n^{\frac{1}{2}} \Sigma_j a^j (\bar{\phi}_n^j - \phi_0^j) \leq y \mid X_{n,1} > \theta_0 + t(cn)^{-1/\alpha} \} \\ & \quad \times \text{pr} \{ X_{n,1} > \theta_0 + t(cn)^{-1/\alpha} \} \\ &= \lim \text{pr} \{ (cn)^{1/\alpha}(\bar{\theta}_n - \theta_0) > t \mid X_{n,1} > \theta_0 + t(cn)^{-1/\alpha} \} \\ & \quad \times \text{pr} \{ n^{\frac{1}{2}} \Sigma_j a^j (\bar{\phi}_n^j - \phi_0^j) \leq y \mid X_{n,1} > \theta_0 + t(cn)^{-1/\alpha} \} \text{pr} \{ X_{n,1} > \theta_0 + t(cn)^{-1/\alpha} \}. \end{aligned}$$

Lemma 7 implies the asymptotic independence of $(cn)^{-1/\alpha}(X_{n,1}-\theta_0)$ and the score statistic evaluated at ϕ_0 for the parameter $\Sigma a^j \phi^j$. It easily follows that the second factor in this expression is independent of t , and hence that the whole expression equals

$$\lim \text{pr} \{ (cn)^{1/\alpha}(\bar{\theta}_n - \theta_0) > t \} \text{pr} \{ n^{\frac{1}{2}} \Sigma_j a^j (\bar{\phi}_n^j - \phi_0^j) \leq y \}.$$

This proves the independence of the asymptotic distributions of $\bar{\theta}_n$ and $\Sigma a^j \bar{\phi}_n^j$ for any $a^1, \dots, a^p$, and hence gives the required result.

Proof of Corollary. Rewrite (4.1) as

$$(nc \log n)^{\frac{1}{2}}(\bar{\theta}_n - \theta_0) \left\{ -\frac{1}{c \log n} \frac{\partial^2 L_n}{\partial \theta^2}(\bar{\theta}_n, \phi_0) \right\}^{\frac{1}{2}},$$

the product of two factors, the first converging to standard normal and the second to 1. This gives (4.1), and (4.2) is similar. Note that (4.1) also holds if $\alpha > 2$.

5. AN ALTERNATIVE TO MAXIMUM LIKELIHOOD

The complicated nature of the preceding results when $1 < \alpha \leq 2$, and the nonexistence of a consistent estimator when $\alpha < 1$, make it desirable to seek some alternative estimator. An obvious candidate for a point estimator of θ is $X_{n,1}$, the sample minimum. The asymptotic distribution of $n^{1/\alpha}(X_{n,1} - \theta_0)$ is Weibull, and Akahira's results show that no point estimator of θ converges at a faster rate when $\alpha < 2$. Thus it seems reasonable to use $X_{n,1}$ as an estimator of θ when $\alpha < 2$. The difficulties are, first, that it is generally not known a priori whether $\alpha < 2$ and, secondly, that we still need an estimator of ϕ .

In this section we propose a new estimator $\tilde{\phi}_n$ of ϕ . It is consistent, so that it may be used to discriminate between the cases $\alpha > 2$, $\alpha < 2$, and when $\alpha < 2$ it is asymptotically efficient, and may therefore be used in place of the maximum likelihood estimator $\hat{\phi}_n$. The new estimator is defined as the local maximum of the function

$$\sum_{k=2}^n \log f(X_{n,k}; X_{n,1}, \phi).$$

It is therefore equivalent to estimating θ by the smallest observation, and dropping that observation for the estimation of ϕ by maximum likelihood.

Define a modified likelihood function

$$\tilde{L}_n(\theta, \phi) = n^{-1} \sum_{k=2}^n \log f(X_{n,k}; \theta, \phi).$$

Thus our new estimator satisfies

$$\partial \tilde{L}_n(X_{n,1}, \tilde{\phi}) / \partial \phi^i = 0 \quad (i = 1, \dots, p). \quad (5.1)$$

THEOREM 4. *Under the conditions of Theorem 1, with probability tending to 1 as $n \rightarrow \infty$, there exists an estimator $\tilde{\phi}_n$ satisfying (5.1), and we have*

$$\tilde{\phi}_n - \phi_n <_{\text{p}} \begin{cases} n^{-1/\alpha} & (\alpha > 1), \\ n^{-1} \log n & (\alpha \leq 1). \end{cases}$$

COROLLARY. *For all α , $\tilde{\phi}_n$ is a consistent estimator of ϕ_0 . If $\alpha < 2$, then $\tilde{\phi}_n$ is also asymptotically efficient, and $n^{\frac{1}{2}}(\tilde{\phi}_n - \phi_0)$ converges to a normal distribution with mean*

zero and covariance matrix M^{-1} , where M is the $p \times p$ matrix with entries

$$m_{ij}(\phi_0) \quad (i, j = 1, \dots, p).$$

Remark. The results of this section continue to hold if $X_{n,1}$ is replaced by any $n^{1/\alpha}$ -consistent estimator $\tilde{\theta}_n$ in (5.1), provided $\tilde{\theta}_n \leq X_{n,1}$.

Proofs. We start with some elementary observations about $\tilde{L}_n$. Clearly,

$$\begin{aligned} L_n(\theta, \phi) - \tilde{L}_n(\theta, \phi) &= n^{-1} \log f(X_{n,1}; \theta, \phi) \\ &= n^{-1} \{\alpha(\phi) - 1\} \log(X_{n,1} - \theta) + n^{-1} \log g(X_{n,1} - \theta; \phi). \end{aligned}$$

When $\theta = \theta_0$ we have

$$\log(X_{n,1} - \theta) = O_p(\log n), \quad \log g(X_{n,1} - \theta; \phi) \rightarrow_p \log c(\phi),$$

so that the whole expression is $O_p(n^{-1} \log n)$. The same applies to

$$(\partial/\partial\phi^i) \{L_n(\theta, \phi) - \tilde{L}_n(\theta, \phi)\},$$

since

$$(\partial/\partial\phi^i) \log g(X_{n,1} - \theta_0; \phi) \rightarrow_p (\partial/\partial\phi^i) \log c(\phi)$$

by Assumption 1. We also note that parts II and III of Lemma 4 apply with $\tilde{L}_n$ in place of L_n , and without the lower bound restriction on $X_{n,1} - \theta_0$.

Let

$$\xi_n^i = (\partial/\partial\phi^i) \tilde{L}_n(\theta_0, \bar{\phi}_n) = (\partial/\partial\phi^i) \{\tilde{L}_n(\theta_0, \bar{\phi}_n) - L_n(\theta_0, \bar{\phi}_n)\}$$

which is $<_p n^{-1} \log n$, by the preceding remarks. Therefore

$$\frac{\partial \tilde{L}_n}{\partial \phi^i}(X_{n,1}, \phi) = \xi_n^i + (X_{n,1} - \theta_0) \frac{\partial^2 \tilde{L}_n}{\partial \theta \partial \phi^i}(\theta^*, \phi^*) + \sum_j (\phi^j - \bar{\phi}_n^j) \frac{\partial^2 \tilde{L}_n}{\partial \phi^i \partial \phi^j}(\theta^*, \phi^*),$$

where $(\theta^*, \phi^*) = \lambda(X_{n,1}, \phi) + (1 - \lambda)(\theta_0, \bar{\phi}_n)$ for some λ ($0 < \lambda < 1$).

Suppose $\alpha \leq 1$. Let $\delta_n, n \geq 1$, be any positive sequence such that $\delta_n \rightarrow 0$, $n\delta_n/\log n \rightarrow \infty$. Define, for $y \in \mathbb{R}^p$ such that $|y| = 1$,

$$f_n(y) = \delta_n^{-2} \tilde{L}_n(X_{n,1}, \bar{\phi}_n + \delta_n y). \quad (5.2)$$

Then

$$\frac{\partial f_n}{\partial y^i} = \delta_n^{-1} \left[\xi_n^i + (X_{n,1} - \theta_0) \frac{\partial^2 L_n}{\partial \theta \partial \phi^i} \{\theta_n^*(y), \phi_n^*(y)\} \right] + \sum_j y^j \frac{\partial^2 L_n}{\partial \phi^i \partial \phi^j} \{\theta_n^*(y), \phi_n^*(y)\},$$

where the dependence of (θ^*, ϕ^*) on both n and y is indicated. Now, for all y with $|y| = 1$, we have

$$|\theta_n^*(y) - \theta_0| < |X_{n,1} - \theta_0| <_p n^{-1/\alpha}, \quad |\phi_n^*(y) - \phi_0| \leq \delta_n \rightarrow 0,$$

so by Lemma 4, modified to apply to $\tilde{L}_n$ as indicated above,

$$\frac{\partial^2 \tilde{L}_n}{\partial \theta \partial \phi^i} \{\theta_n^*(y), \phi_n^*(y)\} <_p \begin{cases} \log n & (\alpha = 1), \\ n^{1/\alpha - 1} & (\alpha < 1), \end{cases}$$

$$\frac{\partial^2 \tilde{L}_n}{\partial \phi^i \partial \phi^j} \{\theta_n^*(y), \phi_n^*(y)\} + m_{ij} \rightarrow_p 0$$

uniformly over y . Therefore

$$\partial f_n / \partial y^i = -\sum_j y^j m_{ij} + e_n^i(y), \quad (5.3)$$

where $e_n^i(y) \rightarrow_p 0$ uniformly over y such that $|y| = 1$. It now follows using Lemma 5, as previously, that with probability tending to 1 there exists $\tilde{\phi}_n$ satisfying $|\tilde{\phi}_n - \bar{\phi}_n| < \delta_n, \partial \tilde{L}_n(X_{n,1}, \tilde{\phi}_n) / \partial \phi^i = 0$ for $i = 1, \dots, p$. Since δ_n may be chosen so that $n\delta_n / \log n$ increases arbitrarily slowly, the result of the theorem follows in the case $\alpha \leq 1$.

The argument for $\alpha > 1$ is almost identical. Now choose δ_n ($n \geq 1$) so that $\delta_n \rightarrow 0, n^{1/\alpha} \delta_n \rightarrow \infty$, and define f_n by (5.2). We have $\partial^2 \tilde{L}_n / \partial \theta \partial \phi^i \rightarrow_p -m_{0i}$ uniformly on a region, so $\delta_n^{-1}(X_{n,1} - \theta_0) <_p \delta_n^{-1} n^{-1/\alpha} \rightarrow 0$ and so (5.3) again holds. The rest of the proof is the same. □

Proof of Corollary. Consistency follows from the relations $\bar{\phi}_n - \phi_0 \rightarrow_p 0, \tilde{\phi}_n - \bar{\phi}_n \rightarrow_p 0$. The results for $\alpha < 2$ follow by noting that

$$n^{\frac{1}{2}}(\tilde{\phi}_n - \phi_0) = n^{\frac{1}{2}}(\bar{\phi}_n - \phi_0) + n^{\frac{1}{2}}(\tilde{\phi}_n - \bar{\phi}_n)$$

and the last term tends in probability to zero. Therefore the asymptotic distribution of $n^{\frac{1}{2}}(\tilde{\phi}_n - \phi_0)$ is the same as that of $n^{\frac{1}{2}}(\bar{\phi}_n - \phi_0)$, which is as claimed.

6. HYPOTHESIS TESTING

In this section we consider testing hypotheses about θ , making first a few remarks about the case ϕ known before turning to the case ϕ unknown. The results given here are of course also relevant to the construction of confidence intervals.

Consider first a simple versus simple test of $H_0: \theta = \theta_0, \phi = \phi_0$ against $H_1: \theta = \theta_n, \phi = \phi_0$, based on sample size n . The Neyman–Pearson test is to reject H_0 if

$$n\{L_n(\theta_n, \phi_0) - L_n(\theta_0, \phi_0)\} > c^* \tag{6.1}$$

for some critical value c^* . If $\alpha < 2$, the natural choice of θ_n is of the form

$$\theta_n = \theta_0 + n^{-1/\alpha} t, \tag{6.2}$$

for fixed real t , since in this case a test of fixed size will have limiting power strictly between 0 and 1.

Now suppose ϕ_0 is unknown but estimated by $\tilde{\phi}_n$, as defined in §5. The obvious analogue of (6.1) is the test: reject H_0 if

$$n\{L_n(\theta_n, \tilde{\phi}_n) - L_n(\theta_0, \tilde{\phi}_n)\} > c^*. \tag{6.3}$$

To compare (6.1) and (6.3), note that

$$\begin{aligned} n\{L_n(\theta_n, \tilde{\phi}_n) - L_n(\theta_0, \tilde{\phi}_n) - L_n(\theta_n, \phi_0) + L_n(\theta_0, \phi_0)\} \\ = n(\theta_n - \theta_0) \sum_j (\tilde{\phi}_n^j - \phi_0^j) (\partial^2 / \partial \theta \partial \phi^j) L_n(\theta_n^*, \phi_n^*) \end{aligned}$$

for some θ_n^*, ϕ_n^* . If $1 < \alpha < 2$ we have

$$\theta_n - \theta_0 = O(n^{-1/\alpha}), \quad \tilde{\phi}_n - \phi_0 <_p n^{-\frac{1}{2}}, \quad \partial^2 L_n / \partial \theta \partial \phi^j <_p 1,$$

so the whole expression is $O_p(n^{\frac{1}{2}-1/\alpha})$ and thus tends to zero in probability. For $\alpha = 1, \alpha < 1$ we have $\partial^2 L_n / \partial \theta \partial \phi^j <_p \log n, n^{1/\alpha-1}$ respectively, leading to the same conclusion. Therefore the two tests (6.1) and (6.3) are asymptotically equivalent, in the sense that they make the same decision in large samples, provided $\alpha < 2$.

Note that this conclusion is false if $\alpha > 2$, even if $\hat{\phi}_n$ is used in place of $\tilde{\phi}_n$. In that case, we have $\theta_n - \theta_0 = O(n^{-\frac{1}{2}})$ and hence that $n(\theta_n - \theta_0) \sum_j (\hat{\phi}_n^j - \phi_0^j) \partial^2 L_n / \partial \theta \partial \phi^j$ is $O_p(1)$ rather than $o_p(1)$, so that the equivalence of (6.1) and (6.3) fails. When $\alpha = 2$, we have

$\tilde{\phi}_n - \phi_0 <_p n^{-\frac{1}{2}}$ by Theorem 4, but in this case $\theta_n - \theta_0$ may be taken to be $O\{(n \log n)^{-\frac{1}{2}}\}$, so that (6.1) and (6.3) are again asymptotically equivalent.

Our conclusion is that for testing a simple hypothesis about θ against a simple alternative, ignorance about ϕ makes no difference to the power of the test when $\alpha \leq 2$, but it decreases the power if $\alpha > 2$.

Now consider a test of $H_0: \theta = \theta_0$ against the composite alternative $H_1: \theta \neq \theta_0$. For the construction of two-sided confidence intervals, in particular, we wish to consider this case. In regular cases, there are three widely used tests, all of which are to first order asymptotically efficient, namely the Wald test which is based on the asymptotic distribution of the maximum likelihood estimator, the score test which is based on the first derivative of the log likelihood and the likelihood ratio test; see, for example, Cox & Hinkley (1974, Ch. 9). In the nonregular case $\alpha < 2$, even when ϕ is known, these tests are not first-order equivalent and it is not known whether any of them has any optimality properties. The following discussion is therefore concerned purely with distributional properties and not with asymptotic optimality.

For the Wald test, the test statistic is the maximum likelihood estimator, assuming it exists. Woodroffe's (1974) results determine the asymptotic distribution of maximum likelihood estimator in the case $1 < \alpha < 2$ but this is not easy to work with. For the likelihood ratio test, nothing appears to be known about the asymptotic distribution of the test statistic. For the score test, however, things are rather simpler, since the asymptotic distribution of the score statistic is known and the test does not require computation or even existence of the maximum likelihood estimator. Therefore we concentrate on the score test, believing that this provides at least a viable method of testing hypotheses about θ even though it may not have asymptotic optimality properties.

The score static for testing $H_0: \theta = \theta_0$ when $\phi = \phi_0$ is known is

$$\partial L_n(\theta_0, \phi_0)/\partial \theta = (\alpha - 1) \Sigma (X_{n,k} - \theta_0)^{-1} n^{-1} + \Sigma \{\partial \log g(X_{n,k} - \theta_0; \phi_0)/\partial \theta_0\} n^{-1}.$$

When $\alpha \leq 2$ the second term is $O_p(1)$ and the first tends to ∞ , but $E(\partial L_n/\partial \theta) = 0$ when $\alpha > 1$, by Assumption 6. Hence $\partial L_n/\partial \theta$ has the same asymptotic distribution as $n^{-1}\{S_{n,1} - E(S_{n,1})\}$ when $\alpha > 1$, $n^{-1}S_{n,1}$ when $\alpha \leq 1$, and this asymptotic distribution is stable, by Lemma 1. In particular, when $1 < \alpha < 2$ or $\alpha < 1$, $n^{1-1/\alpha} \partial L_n/\partial \theta$ has a nondegenerate stable limit law.

A two-sided test is therefore defined by the acceptance region

$$a < n\{a_n^{-1} \partial L_n(\theta, \phi_0)/\partial \theta - b_n\} < b, \quad (6.4)$$

where a_n, b_n are as in Lemma 1 and the percentage points a, b are determined by the limiting stable law. For $\alpha = 2$, the acceptance region is

$$a < n^{\frac{1}{2}}(c \log n)^{-\frac{1}{2}} \partial L_n(\theta, \phi_0)/\partial \theta < b \quad (6.5)$$

with a, b the appropriate percentage points of the standard normal distribution.

Now let us consider the effect of ϕ being unknown, and estimated by $\tilde{\phi}_n$. When $1 < \alpha < 2$ we have

$$n^{1-1/\alpha} \{\partial L_n(\theta, \tilde{\phi}_n)/\partial \theta - \partial L_n(\theta, \phi_0)/\partial \theta\} = n^{1-1/\alpha} \Sigma_j (\tilde{\phi}_n^j - \phi_0^j) \partial^2 L_n(\theta, \phi_n^*)/\partial \theta \partial \phi^j,$$

with ϕ_n^* between $\tilde{\phi}_n$ and ϕ_0 . But $\partial^2 L_n/\partial \theta \partial \phi^j <_p 1$ uniformly on a region of the form $|\phi - \phi_0| = o_p(1)$, $|\theta - \theta_0| = o_p(n^{-2/\alpha+1})$, so the whole expression $\rightarrow_p 0$ uniformly on a region of form $|\theta - \theta_0| = O(n^{-1/\alpha})$. Similar arguments show that the same result holds for

$0 < \alpha < 1$, $\alpha = 1$ and $\alpha = 2$. We therefore conclude that the tests (6.4)–(6.5) remain valid when ϕ_0 is unknown, provided that ϕ_0 is replaced by its estimated value $\tilde{\phi}_n$ or by some other $\sqrt{n}$ -consistent estimator.

As previously remarked, no claim of optimality is made for these procedures. In one-parameter problems, θ unknown, ϕ known, the problem of constructing asymptotically optimal procedures, when $\alpha < 2$, is considered in detail in Chapters 5 and 6 of Ibragimov & Has'minskii (1981). These results are complicated and depend on the loss function.

7. APPLICATIONS IN EXTREME VALUE THEORY

The results of this paper may be applied to two particular distributions which are of importance in the analysis of extreme values. These are the generalized extreme value distribution, which includes the three-parameter Weibull as a special case, and the generalized Pareto distribution introduced by Pickands (1975).

The density function of the generalized extreme value distribution is

$$g(y; k, \mu, \sigma) = \sigma^{-1} \{1 - k\sigma^{-1}(y - \mu)\}^{1/k-1} \exp[-\{1 - k\sigma^{-1}(y - \mu)\}^{1/k}] \quad (7.1)$$

defined on the set $\{y: k(y - \mu) < \sigma\}$. In the case $k > 0$, this is a reparameterization of the Weibull distribution, while the cases $k < 0$, $k = 0$, defined by taking the limit as $k \rightarrow 0$ in (7.1), correspond to the type II and type I extreme value laws, in Gumbel's (1958) characterization. For computational details, see Prescott & Walden (1980, 1983). Recent reviews on the fitting of extreme value distributions and discrimination among types have been given by Mann (1984) and Tiago de Oliveira (1984).

For $k > 0$, the results of this paper are directly applicable, and show in particular that the classical properties of maximum likelihood estimators hold for $0 < k < \frac{1}{2}$, but not for $k \geq \frac{1}{2}$. The information matrix is finite over the whole range $-\infty < k < \frac{1}{2}$. For $k < 0$, the range of the distribution again depends on the unknown parameters, the density being positive when $y > \mu + \sigma/k$. Writing $\beta = -1/k$, $\theta = \mu + \sigma/k$ gives the reparameterization

$$g(y; \beta, \theta, \sigma) = \sigma^{-1} \{(y - \theta)/(\beta\sigma)\}^{-\beta-1} \exp[-\{(y - \theta)/(\beta\sigma)\}^{-\beta}] \quad (y > \theta),$$

which converges to zero faster than any power of $y - \theta$ as $y \downarrow \theta$. Although this does not fall within the scope of our Theorem 1, the arguments that were applied there to the case $\alpha > 2$ are applicable here also, and show the existence of a maximum likelihood estimator which is asymptotically normal and efficient. We conclude that the classical asymptotic properties of maximum likelihood estimators hold throughout the range $-\infty < k < \frac{1}{2}$, while for $k \geq \frac{1}{2}$ the results for the three-parameter Weibull are directly applicable.

We remark that this argument applies also to the three-parameter log normal distribution. In this case also, the likelihood function is unbounded but there is a local maximum which is asymptotically normal and efficient. This observation provides a theoretical justification for the procedures advocated by Griffiths (1980). Another distribution for which similar arguments hold is the inverse Gaussian distribution with unknown location (Cheng & Amin, 1981).

The generalized Pareto distribution has density

$$g(y; k, \sigma) = \sigma^{-1} (1 - ky/\sigma)^{1/k-1}, \quad (7.2)$$

defined on the set $y > 0$, $ky < \sigma$. Its importance in extreme value theory arises from the fact that it characterizes the limiting distributions of the excesses over a threshold, as the threshold tends to infinity (Pickands, 1975; Davison, 1984; Smith, 1984). When

$k < 0$, it is the same as the Pareto distribution, and the limit $k \rightarrow 0$ gives the exponential distribution with mean σ . For $k > 0$, the density is a transformation of the three-parameter beta distribution of § 1, with $\beta = 1$. Thus we get analogous results to those for the generalized extreme value distribution: for $k < \frac{1}{2}$ the information matrix is finite and the classical asymptotic theory of maximum likelihood estimators is applicable, while for $k \geq \frac{1}{2}$ the problem is nonregular and special procedures are needed.

An alternative approach was taken by Hall (1982), who considered the estimation of θ when the density is of the form (1.1), but without assuming a specific parametric form for f_0 . Hall proposed a procedure based on a specified number of order statistics; his procedure is almost the same as restricting attention to observations beneath a threshold, and assuming the generalized Pareto distribution for the differences between these observations and the threshold. Hall proves asymptotic normality of his estimators in the case $\alpha \geq 2$. The strength of his approach is that it also takes into account the error in (1.1), but he essentially assumes that this error is known, which is not a realistic practical assumption.

In conclusion, there are many open questions. Even when $\alpha > 2$, not all the higher-order moments of the score statistic are finite and it remains an open question whether arguments based on higher-order asymptotics, e.g. Efron & Hinkley (1978), Barndorff-Nielsen (1983), are applicable. Mann (1984) has reviewed a number of papers in which authors have reported practical difficulties in estimating the parameters of the three-parameter Weibull. This raises questions of the speed of convergence, and small-sample properties of the estimators, which we have not considered at all. The problem of determining asymptotically efficient tests and estimators of θ when $\alpha < 2$ is also unresolved. Finally, there is the question of the asymptotic performance of Bayes estimators; in the case $\alpha < 2$ we would not expect asymptotic normality (Dawid, 1970) to hold, and it would be of interest to establish under what conditions the asymptotic posterior distribution does not depend on the prior distribution.

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